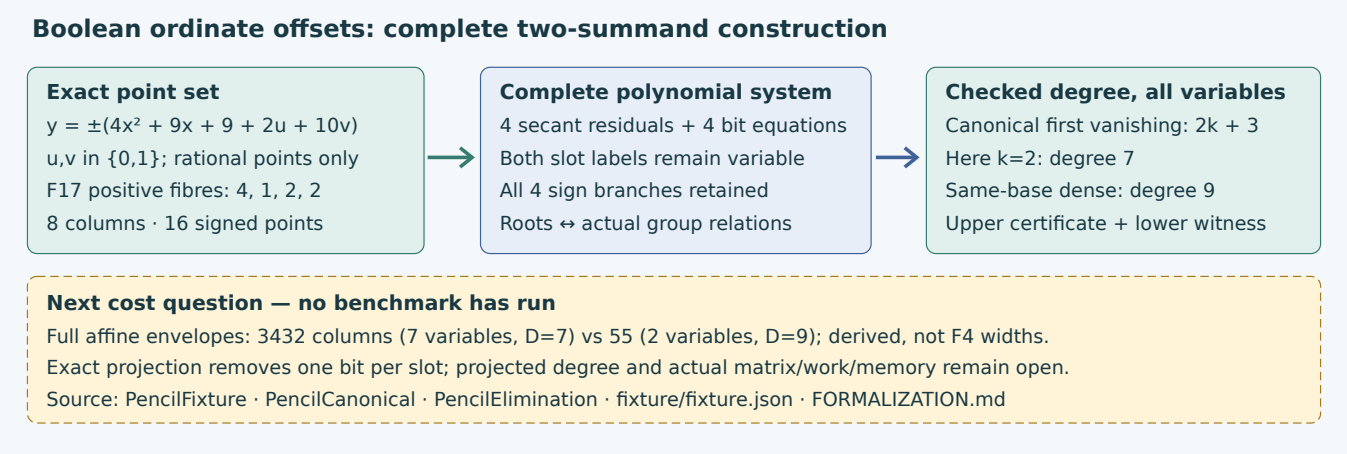
# Boolean offsets of a quadratic ordinate base

2026-11-01 · construction and checked degree continuation

A complete two-summand family now has a Lean-checked canonical homogeneous degree law: **2k+3**, including all 2k Boolean selector variables. The factor base contains every rational point with $y=\pm(a\,x^2+b\,x+c+t0+\mathrm{sum}(w_i u_i))$, $u_i^2-u_i=0$. Both slots' labels remain variables; individual shifted graphs need not split completely. Empty fibres supply no points.




## A larger exact fixture

Curve icv1-fp5-tm1-a77cc6af is $y^2=x^3+2x+2$ over F17. Use $h0=4x^2+9x+9$ and weights [2,10]. The offsets {0,2,10,12} give positive fibre sizes [4,1,2,2]. Eighteen signed labels deduplicate to **16 signed points at 8 abscissae**: {0,3,5,6,7,9,10,16}, twice the old fixture's sizes. The new base is FB1h29cc024e755c. Its full selector has canonical degree **7**, versus **9** for the same-base dense membership/S3 presentation.

Lean proves exact finite membership and cardinalities, actual polynomial root/group equivalence, and raw-root soundness without assumed membership. All signs, tangents and repeated abscissae are retained. Infinity has a separate exact inverse-pair theorem. Metadata remains unregistered; group order is inherited, not newly proved.

This is a hand-derived fixture selected after viewing the older F17 example, not a prospective sample or natural-yield measurement. Duplicate labels at x=9 are not extra columns. No external novelty is established.

# Degree certificate, variable tax and next experiment

The actual highest forms normalize to the core ($m^2$, $xz$, $x^2-\text{alpha}\cdot mx$, $z^2-\text{beta}\cdot mz$) plus one square for every Boolean variable. Every degree $2k+3$ monomial either repeats a bit or has core degree at least 3, so it vanishes. A degree $2k+2$ witness $m\cdot x\cdot\text{product(bits)}$ survives via a map to a pure-square monomial quotient. The hypotheses are nonzero quadratic coefficients and characteristic other than 2; the target and all lower coefficients are arbitrary. FORMALIZATION.md maps claims to exact signatures.

## The remaining cost comparison

The seven-variable degree 7 full affine monomial envelope is choose(14,7)=3432 columns; the two-variable dense degree 9 envelope is choose(11,2)=55. These are **derived ambient budgets**, not measured matrix widths. Lower canonical degree alone does not establish lower complete solving work.

Exact Boolean projection is proved even for zero weight: exists u in {0,1}: $d+l\cdot u=0$ iff $d(d+l)=0$. Eliminating one bit per slot removes two variables and gives quartic residuals. Its actual canonical degree and solving cost still need checking. The universal elimination identity is kernel checked separately from evaluated-tier IDC1hb79cfe74b833e0be.

## Validation and execution status

The unchanged transitive axiom audit passes **240 declarations in 18 modules**, with zero failures and zero native trust. All five new modules pass warnings-as-errors. Thirteen native unit tests, formatting and clippy pass; the deterministic FB1/fixture/identity export replays byte-for-byte. Lean 4.19.0 and nine Git dependencies are pinned and checked. Forty source inputs are hashed. Compiler failures and accepted source are preserved. lean4checker is unavailable; no fresh checker pass is claimed.

No new timing, affine F4 degree or relation-yield measurement has run. PROTOCOL.json records the matched panel; its full runner and controls are not yet implemented. EKS arbitrary-command submission is not configured here, and the inspected GitHub path accepts ecbench specifications. The old local exception is not reused. Original P-256/17-term gates remain open.

## Sources and replay

See reports/REPORT.md, FORMALIZATION.md and verification/final/ for exact scope, hypotheses and receipts. Run `bash verify.sh NEW_RECEIPTS_DIR`. The prior 24-design register and atlas already cover learned coordinates, digits and generic towers. Primary PKM 2016: https://people.maths.ox.ac.uk/petit/files/16PKC_primeECDLP.pdf. Actual group laws reuse Mathlib and Angdinata–Xu, ITP 2023: https://doi.org/10.4230/LIPIcs.ITP.2023.6.